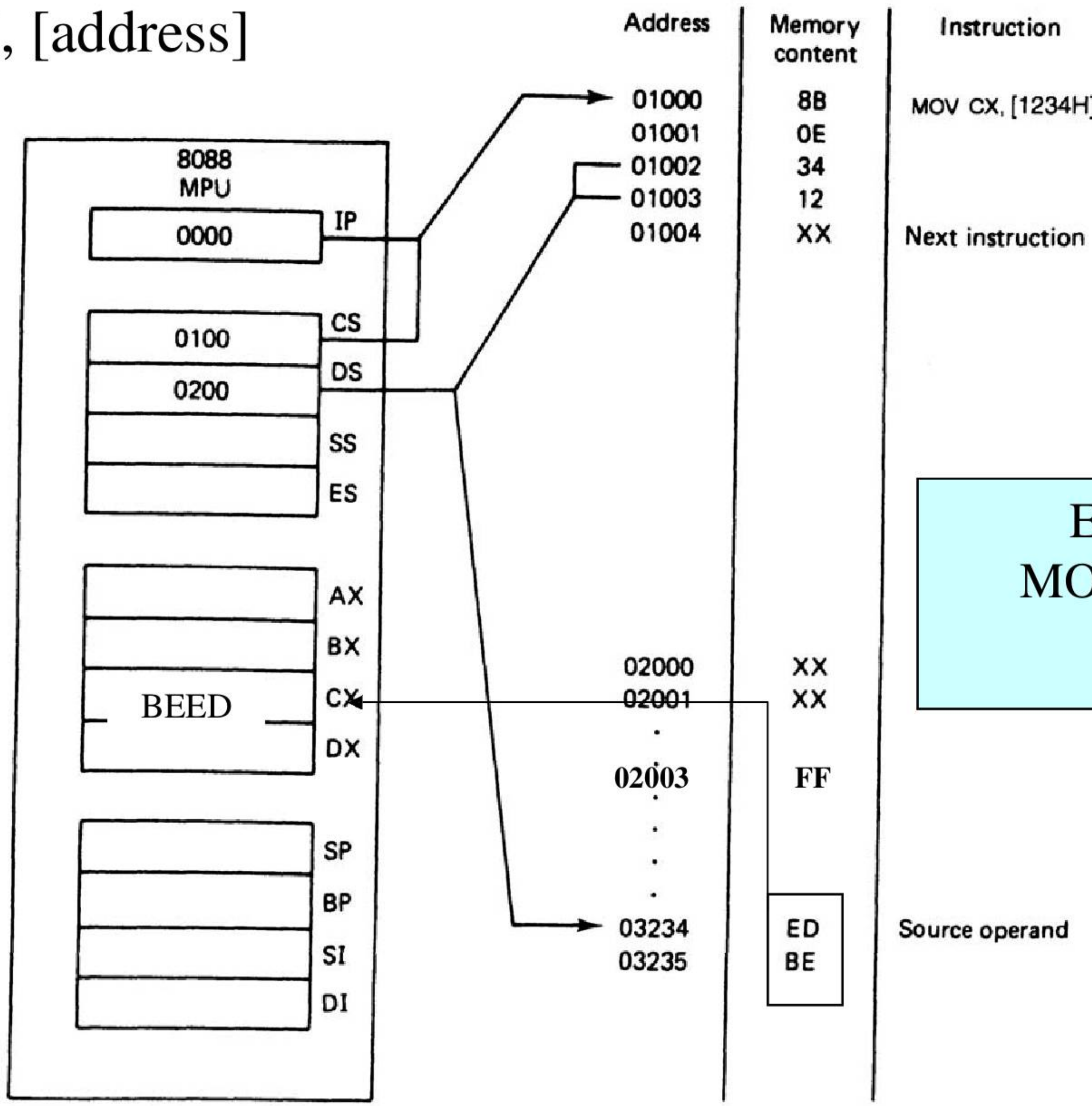
Addressing Modes

- When the 8088 executes an instruction, it performs the specified function on data
- These data, called <u>operands,</u>
  - May be a part of the instruction
  - May reside in one of the internal registers of the microprocessor
  - May be stored at an address in memory
- **Register Addressing Mode**
  - MOV AX, BX
  - MOV ES,AX
  - MOV AL,BH
- **Immediate Addressing Mode**
  - MOV AL,15h
  - MOV AX,2550h
  - MOV CX,625

Direct Addressing Mode

MOV CX, [address]

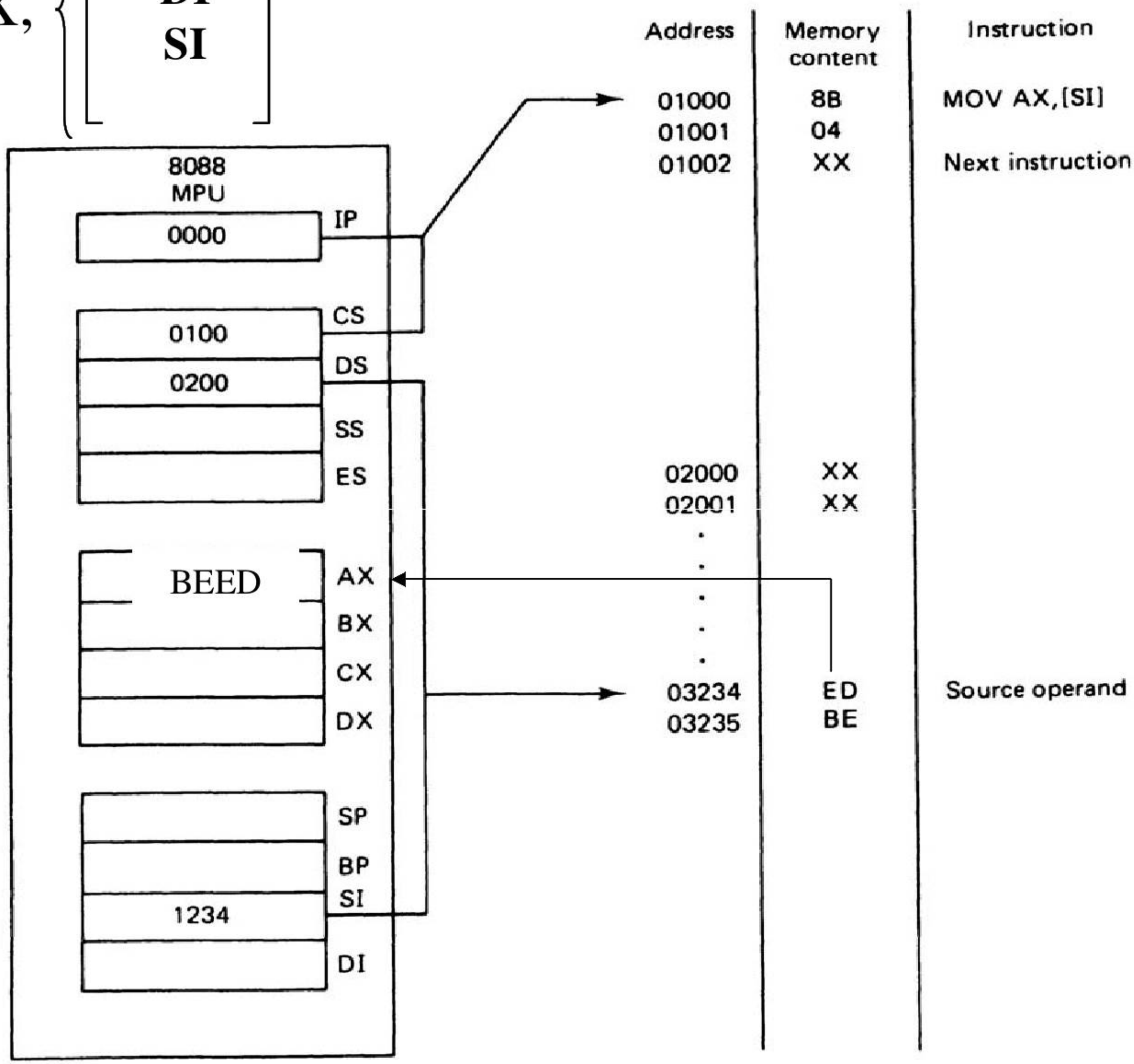


Example:
MOV AL,[03]

AL=?

Register Indirect Addressing Mode

MOV AX, [BX / DI / SI]



Example for Register Indirect Addressing

- Assume that DS=1120, SI=2498 and AX=17FE show the memory locations after the execution of:

MOV [SI],AX

DS (Shifted Left) + SI = 13698.

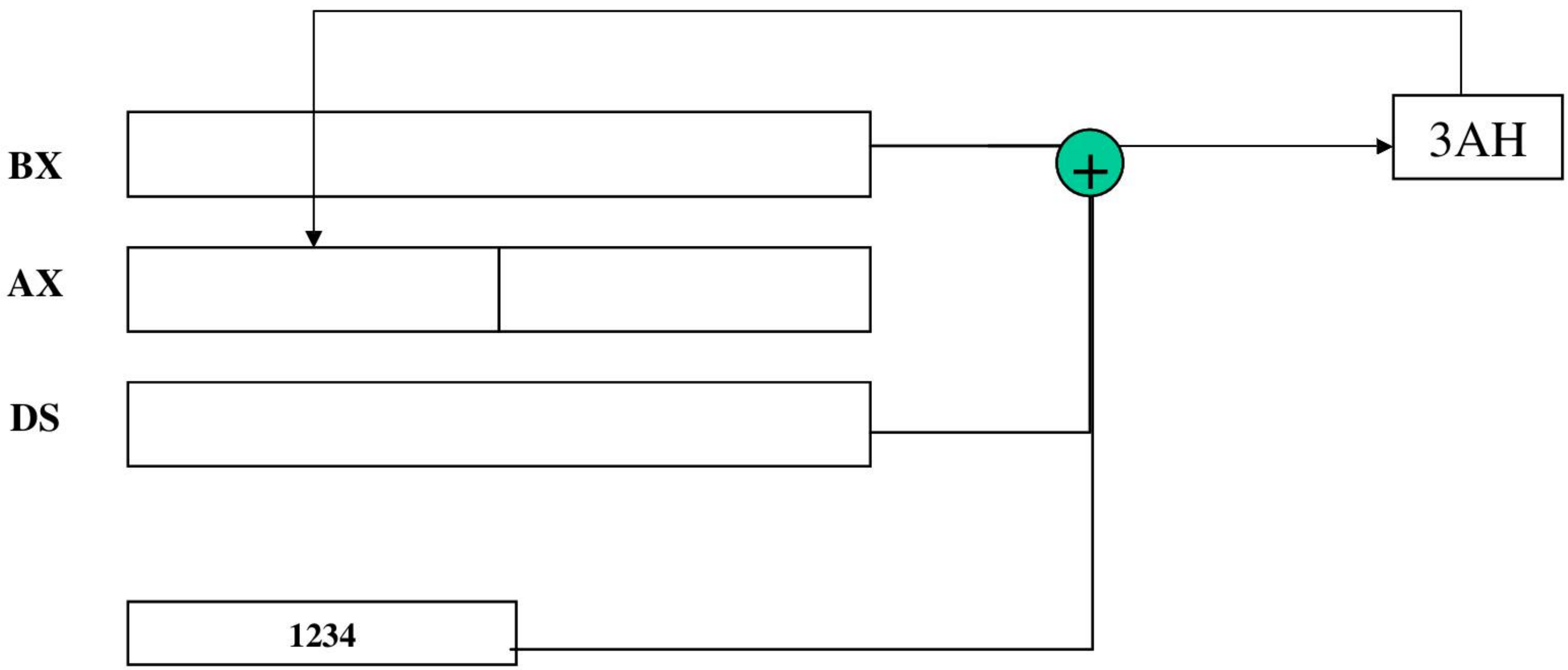
With little endian convention:

Low address 13698 → FE

High Address 13699 → 17

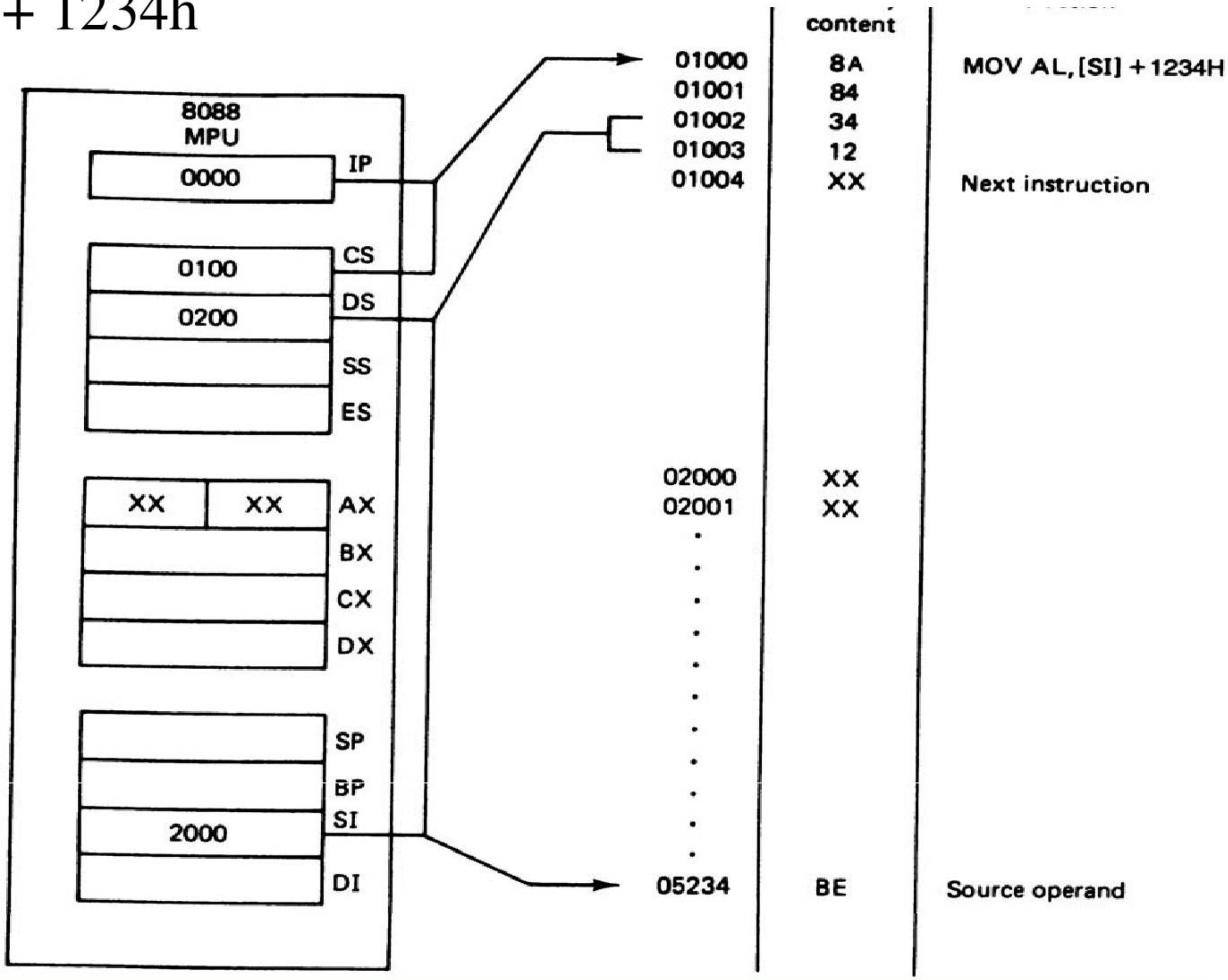
Based-Relative Addressing Mode

MOV AH, [DS:BX / SS:BP] + 1234h



Indexed Relative Addressing Mode

MOV AH, [$\genfrac{}{}{0pt}{}{SI}{DI}$] + 1234h



Example: What is the physical address MOV [DI-8],BL if DS=200 & DI=30h ?
DS:200 shift left once 2000 + DI + -8 = 2028

Based-Indexed Relative Addressing Mode

- Based Relative + Indexed Relative
- We must calculate the PA (physical address)

$$PA = \boxed{\begin{array}{c} CS \\ SS \\ DS \\ ES \end{array}} : \boxed{\begin{array}{c} BX \\ BP \end{array}} + \boxed{\begin{array}{c} SI \\ DI \end{array}} + \boxed{\begin{array}{c} \text{8 bit displacement} \\ \text{16 bit displacement} \end{array}}$$

$$\left.\begin{array}{c} \text{MOV AH,[BP+SI+29]} \\ \text{or} \\ \text{MOV AH,[SI+29+BP]} \\ \text{or} \\ \text{MOV AH,[SI][BP]+29} \end{array}\right\} \begin{array}{c} \text{The} \\ \text{register} \\ \text{order does} \\ \text{not matter} \end{array}$$

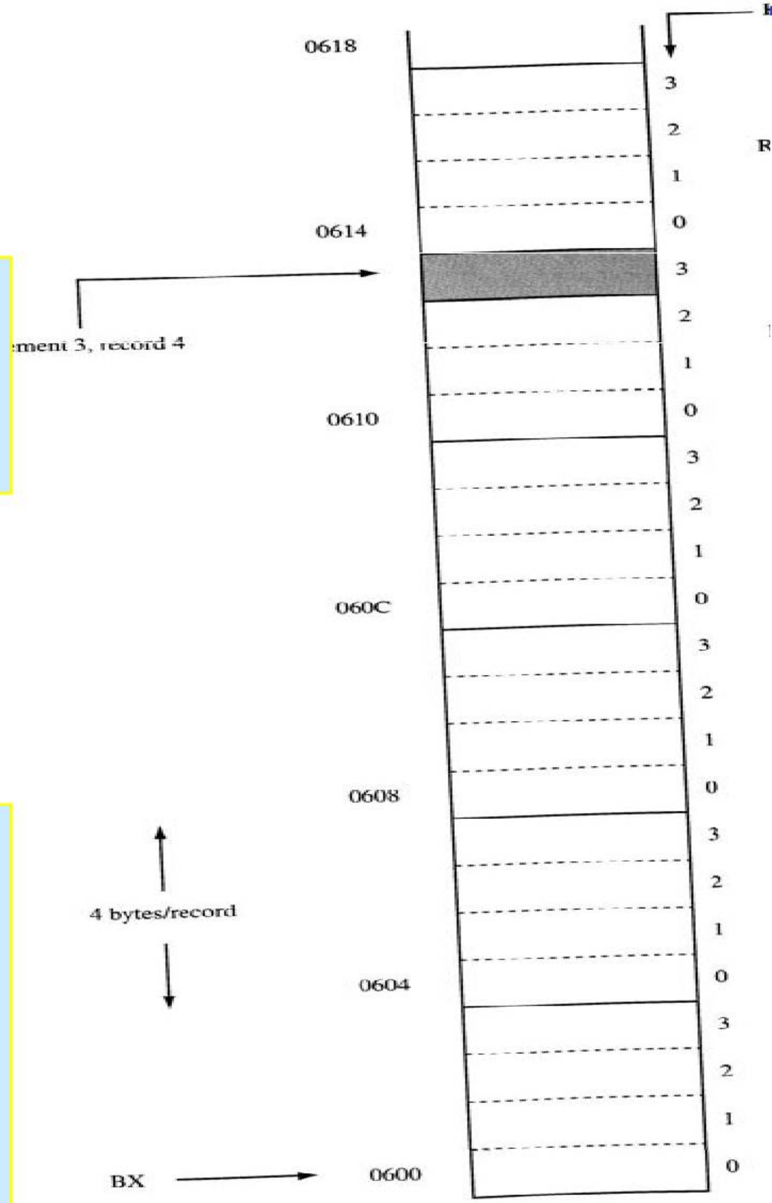
Based-Indexed Addressing Mode

ure 4–4.
e base + index +
placement
dressing mode can
used to access a
rticular element in a
rticular record of an
ay.

```
MOV BX, 0600h
MOV SI, 0010h ; 4 records, 4 elements each.
MOV AL, [BX + SI + 3]
```

OR

```
MOV BX, 0600h
MOV AX, 004h ;
MOV CX,04;
MUL CX
MOV SI, AX
MOV AL, [BX + SI + 3]
```



Summary of the addressing modes

Addressing Mode	Operand	Default Segment
Register	Reg	None
Immediate	Data	None
Direct	[offset]	DS
Register Indirect	[BX] [SI] [DI]	DS DS DS
Based Relative	[BX]+disp [BP]+disp	DS SS
Indexed Relative	[DI]+disp [SI]+disp	DS DS
Based Indexed Relative	[BX][SI or DI]+disp [BP][SI or DI]+disp	DS SS

16 bit Segment Register Assignments

Segment Registers	CS	DS	ES	SS
Offset Register	IP	SI,DI,BX	SI,DI,BX	SP,BP

Type of Memory Reference	Default Segment	Alternate Segment	Offset
Instruction Fetch	CS	none	IP
Stack Operations	SS	none	SP,BP
General Data	DS	CS,ES,SS	BX, address
String Source	DS	CS,ES,SS	SI, DI, address
String Destination	ES	None	DI

Segment override

Instruction Examples	Override Segment Used	Default Segment
MOV AX,CS:[BP]	CS:BP	SS:BP
MOV DX,SS:[SI]	SS:SI	DS:SI
MOV AX,DS:[BP]	DS:BP	SS:BP
MOV CX,ES:[BX]+12	ES:BX+12	DS:BX+12
MOV SS:[BX][DI]+32,AX	SS:BX+DI+32	DS:BX+DI+32

Example for default segments

- The following registers are used as offsets. Assuming that the default segment used to get the logical address, give the segment register associated?

a) BP b)DI c)IP d)SI, e)SP, f) BX

- Show the contents of the related memory locations after the execution of this instruction

  MOV [BP][SI]+10,DX

  if DS=2000, SS=3000,CS=1000,SI=4000,BP=7000,DX=1299 (all hex)

SS(0)=30000

30000+4000+7000+10=3B010